

PRE-BUILD RESEARCH DOSSIER

Ammonia Plant Builder Agent

A stress test of the architecture, a map of the competitive white space, a hardened engineering core with verified correlations and validation sources, and a staged roadmap from solid MVP to an extremely interactive plant trainer.

Prepared for Daemon · KNUST Chemical Engineering

Research pass · August 2026 · Scope: ammonia only

CORRECTIONS · LANDSCAPE · ENGINE · INTERACTIVITY ·
STACK · RISKS · ROADMAP

Table of Contents

1. Executive Summary and Verdict	3
1.1 The verdict on the current plan	3
1.2 Five corrections to make before writing code	3
1.3 The position this product can own	4
2. Stress Test: Where the Plan Holds and Where It Breaks	5
2.1 What the plan gets right	5
2.2 Seven engineering corrections	6
2.3 Architecture gaps the plan does not address	7
2.4 Layer-by-layer scorecard	8
3. The Landscape: What Exists, What Is Missing	9
3.1 The cost wall: commercial simulators	9
3.2 Free engines: the mechanics are already commoditized	9
3.3 The AI wave: research is arriving, products are text-only and pro-priced	10
3.4 The training-simulator world: what the pros pay	10
3.5 The ammonia demand signal	11
3.6 White space and threats	11
4. The Hardened Engine: Correlations, Data, Validation	12
4.1 The stream object, finalized	12
4.2 Unit-operation modeling table	12
4.3 The ammonia equilibrium constant, implementation-ready	13
4.4 Convergence strategy for the synthesis loop	15
4.5 Validation plan: what "correct" means, numerically	15
4.6 Energy numbers worth hard-coding	16
5. The Interactivity Frontier	16
5.1 The interactivity ladder	16
5.2 Frontier A — live conversational voice	17
5.3 Frontier B — the 3D spatial plant	19
5.4 Frontier C — gamified operations	20
5.5 The design science underneath	20

6. Recommended Technology Stack	21
6.1 The stack, layer by layer	21
6.2 TTS economics, verified	22
6.3 The LLM integration pattern, specified	23
6.4 Cost model at three scales	24
7. Risk Register	24
8. Roadmap v2: Stages, Effort and the Critical Path	26
8.1 The stages	26
8.2 What changes from the original plan, and why	27
8.3 The critical path: protect the what-if loop	27
9. Validation Source Library	28
9.1 Engineering anchors, ranked	28
9.2 How to use the library	29
9.3 Key web sources for the stack and landscape chapters	30

1. Executive Summary and Verdict

This dossier is the pre-build research pass over the ammonia plant builder agent plan. It stress-tests the architecture, maps the competitive landscape, hardens the engineering layer with verified correlations and data, explores how far the interactivity can be pushed, and re-cuts the roadmap with realistic effort estimates. Every claim in this document traces to either a live web source found during research (August 2026) or a published engineering reference; the full source library is ranked in Chapter 9.

1.1 The verdict on the current plan

The plan's central bet is correct, and it is worth saying plainly: **the physics/language separation is the right architecture, and it is exactly where the industry is heading.** GSE Solutions, the largest operator-training-simulator vendor, announced a 2026 collaboration with Nucleon to author training scenarios from natural-language prompts on top of deterministic simulation — the same LLM-as-parameterizer, engine-as-calculator pattern the plan describes. Siemens shipped a natural-language copilot inside NX CAD in 2025. OpenAI's structured outputs feature reports 100% JSON-schema adherence versus roughly 40% for prompt-only JSON, which validates the plan's decision to make intake a single constrained extraction call rather than a free-form agent loop. The plan is not just defensible; it is early.

The scope narrowing to ammonia only is also confirmed as the right call. Research found no free, full-flowsheet ammonia simulator anywhere — not in education, not in open source, not from vendors. The closest free artifact is a Flash-era equilibrium toy from Annenberg Learner. A bounded nine-unit-operation library that models the complete natural-gas-to-NH₃ train would be the first of its kind, positioned in front of a documented demand wave: the green-ammonia build-out and a workforce gap measured in tens of thousands of workers by 2030.

1.2 Five corrections to make before writing code

The stress test in Chapter 2 surfaces issues that would cost real credibility if discovered later by a reviewing engineer. None of them changes the architecture; all of them change the first month of implementation.

#	Correction	Why it matters
C1	Converter approach-to-equilibrium is 90–95% per bed, not the 60–70% the plan assumes. Industry design convention (Cheema and Krewer 2018; Topsoe/Casale practice) holds each bed outlet at 90–95% of equilibrium. At 0.65 the model under-predicts converter outlet NH_3 by 3–5 percentage points.	The synthesis loop is the flagship of the whole plant; its numbers must survive an engineer's scrutiny on first demo.
C2	Neither CoolProp nor Cantera can compute the $\text{NH}_3\text{--H}_2\text{--N}_2\text{--CH}_4\text{--Ar}$ flash at -20°C and 150 bar. CoolProp's mixture framework is GERG-style and NH_3 is not a GERG component; Cantera has no real-fluid liquid flash. Use the <i>thermo</i> library (MIT) with Peng–Robinson and binary interaction parameters.	The plan currently cites Cantera/CoolProp for thermophysics; the condenser model would be built on a library that cannot do its job.
C3	Model the secondary reformer as two zones: adiabatic combustion of process air, then catalytic equilibrium. Air flow is the degree of freedom that sets $\text{H}_2/\text{N}_2 = 3$.	A single equilibrium block cannot close the nitrogen balance, and the nitrogen balance defines the plant.
C4	Carry argon explicitly (0.93% of air) through the whole flowsheet.	Purge sizing and loop inert levels (10–15 mol%) depend entirely on Ar; dropping it breaks the purge story.
C5	Validate against EFMA, Flórez-Orrego (2017), the 2024 TAMU thesis and the Rice CENG403 design projects. Turton and Felder contain no ammonia case study; "Temperley" does not exist; "Maxwell's thesis" is unverifiable.	The validation anchors named in the plan will not actually validate an ammonia plant.

Table 1.1 — The five must-fix corrections, in the order they should be applied.

1.3 The position this product can own

Chapter 3 maps the terrain: commercial simulators sit behind a \$10k–100k/year cost wall, free simulators have no pedagogy, AI copilots are text-only and professional-priced, and industrial operator training costs \$800k–\$3M per installation. The verified white space is the intersection nobody occupies:

The free, voice-narrated, OTS-style ammonia plant trainer. One deeply modeled plant, explained out loud, recomputed live for every question a student asks, exportable as a real mass balance. Mechanics are commoditized; nobody owns pedagogy + voice + ammonia + 3D together.

The headline numbers behind that position, all sourced in later chapters:

Number	Value	Source chapter
Commercial simulator license wall	\$10,000–\$100,000+ / year	Ch. 3
Industrial OTS installation cost	\$800,000–\$3,000,000	Ch. 3
Existing free ammonia simulators	1 (equilibrium-only toy)	Ch. 3
Narration TTS cost, 100 walkthroughs	\$4.50 — or \$0 on Azure free tier	Ch. 6
Achievable what-if voice latency	0.4–0.5 s chunked; 0.5–0.8 s full-duplex	Ch. 5
BAT specific energy, SMR ammonia	28–29 GJ/t NH ₃	Ch. 4
Solo build, part-time, to v3 operator mode	≈22 working weeks	Ch. 8

Table 1.2 — Headline research numbers.

The single most important strategic finding, echoing and sharpening the plan’s own instinct: the gap between an impressive demo and a credible product is entirely whether the what-if loop is real. A reviewer’s first test will be to change a number and check whether the plant recomputes or improvises. Everything in this dossier ultimately serves that test.

2. Stress Test: Where the Plan Holds and Where It Breaks

This chapter takes the plan’s thirteen sections one by one and asks what survives contact with evidence. The structure is deliberately asymmetric: strengths get a short confirmation, problems get the detail, because problems are where pre-build research pays for itself.

2.1 What the plan gets right

- 1. The physics/language split.** Every adjacent signal validates it: GSE + Nuclearn (2026) authoring OTS scenarios with LLMs over deterministic simulation; Siemens Design Copilot in NX; the AutoChemSchematic research agent that verifies DWSIM material balances. The pattern "LLM emits tool calls, deterministic engine computes, LLM narrates the diff" is where serious engineering AI converged in 2025–2026.
- 2. Bounded ammonia scope.** The library is genuinely finite — roughly a dozen models including the desulfurizer and circulator the plan omits — and every one has published operating data behind it (Chapter 4). No competitor has built the free version of any of it.
- 3. The build order.** Testing plumbing with a pure-mass-balance splitter before any physics, then the closed-form compressor, then once-through reactors, and only then the recycle loop, is exactly how seven of nine unit operations get validated with plain sequential math. This de-risks the one genuinely hard numerical problem.

4. **Warm-started what-ifs.** Correct, and stronger than the plan states: with a good tear initialization, a small perturbation typically converges in a handful of iterations (Chapter 4 gives the strategy and the iteration budget), which is what makes a conversational recompute loop feel instant.
5. **Templated narration for structured deltas.** Filling a template from real before/after numbers costs zero LLM calls and cannot hallucinate — this remains one of the sharpest design decisions in the plan.

2.2 Seven engineering corrections

Each correction below is stated with its evidence. C1–C5 also appear in the executive summary; here they are given with the numbers an implementer needs.

C1 — Converter approach to equilibrium: 0.90–0.95, not 0.60–0.70

The plan's "60–70% fractional approach" reads like a conservatism choice, but it is outside industrial design practice. Published converter models hold each bed at 90–95% of equilibrium conversion, or equivalently a temperature approach of 15–30 °C; Cheema and Krewer's power-to-ammonia design (RSC Advances, 2018) pins bed outlets at 90% of the equilibrium temperature, and the Flórez-Orrego industrial model (Energy, 2017) agrees. At 450 °C and 200 bar the verified equilibrium is 27.4 mol% NH_3 ; a 0.65 fractional approach predicts 17.8 mol% where industry expects 24.7–26.1 mol% — an error large enough to poison every downstream number in the loop, including product rate, recycle ratio and purge. Chapter 4's Figure 4.1 makes the gap visible.

C2 — The thermophysical library for the condenser is wrong

The plan lists Cantera and CoolProp as the property layer. For reactor equilibrium they are usable (Cantera's Gibbs minimization is a good cross-check). For the synthesis-loop condenser they are not: CoolProp's mixture framework is a GERG-2008-style multi-fluid model, NH_3 is not among its supported mixture components (the NH_3 –water pair is explicitly unavailable, GitHub issue #341), and mixture flashes support only a small set of pre-generated operations (issue #2534, April 2025). Cantera performs no real-fluid liquid flash at all. The industry-standard answer for NH_3 -containing loop mixtures is a cubic equation of state — Peng–Robinson with binary interaction parameters — which is precisely what Caleb Bell's *thermo* library (MIT) provides, in pure Python, with flash algorithms and mixture enthalpies. Research-grade reference EOS exist for exactly the loop binaries (Herrmann et al., Fluid Phase Equilibria, 2020) and serve as the validation target. Recommended split: *thermo* for the loop VLE and enthalpies, CoolProp for pure NH_3 and IF97 steam, hand-coded K_p correlations for reactors.

C3 — The secondary reformer is two zones, not one equilibrium block

Process air enters the secondary reformer and its oxygen burns hydrogen (fastest consumer) in an adiabatic combustion zone that lifts the gas to roughly 950–1000 °C; the residual methane then approaches equilibrium over the catalyst bed, exiting at 0.2–0.3% dry CH_4 . Modeling this as a single equilibrium reactor cannot represent the combustion step, and more importantly the air flow is the design degree of freedom that fixes the final H_2/N_2 ratio at 3 — the specification the whole plant serves. The two-zone recipe with air as DOF is in Chapter 4, Table 4.2.

C4 — Argon must be an explicit species

Air brings 0.93% argon. Ar is inert, accumulates in the loop, and together with methane sets the inert fraction (industry: 10–15 mol%) that the purge exists to control. A species set without Ar produces a loop that never needs its purge — silently breaking one of the most instructive stories the narrator can tell. The stream schema in Chapter 4 carries nine species including Ar.

C5 — Citation hygiene: three anchors do not exist

The plan's open items mention Gillespie–Beattie (correct — verified below) but the research also surfaced three phantom references in the surrounding material: "Temperley" is not a real correlation for ammonia synthesis equilibrium (the lineage is Larson–Dodge, 1923, then Gillespie–Beattie, 1930, then the textbook tables in Vancini 1971 and Appl 1999); "Maxwell's thesis" could not be located in any library or repository; and Turton's and Felder's textbooks, while canonical for process design generally, contain no complete ammonia-plant case study with a stream table. Validation must anchor on the real sources ranked in Chapter 9.

C6 — Gillespie–Beattie needs its pressure-dependent coefficient table

The correlation is the right choice and was numerically re-verified during this research (classic values reproduce exactly: 16.3% NH_3 at 450 °C/100 bar; 38.2% at 400 °C/200 bar; 26.4% at 500 °C/300 bar). But it is an *apparent* equilibrium constant in partial-pressure terms whose coefficients β and I are themselves pressure-dependent — a single pressure-independent formula is wrong across the 100–300 bar range this plant spans. The full implementation-ready table, units and validity limits are given in Chapter 4, Section 4.3.

C7 — Convergence needs a strategy, not just a method

The plan names direct substitution or Wegstein, which is a reasonable start but incomplete. Wegstein's acceleration factor must be bounded (commercial simulators clip it to $[-5, 0]$) or it overshoots and oscillates; the tear stream must be initialized close to the answer or iterations are wasted and the equilibrium solve can find the trivial no-reaction root. The hardened recipe — damped direct substitution for three iterations, then bounded Wegstein, Broyden as the stall fallback, and specific tear initialization values — is Section 4.4.

2.3 Architecture gaps the plan does not address

- **No validation layer between the LLM and the engine.** Structured outputs guarantee the JSON shape, not the values: an LLM can emit 900 bar loop pressure in perfectly valid JSON. A Pydantic model with hard physical ranges (loop 80–250 bar, S/C 2–4, bed temperatures 350–530 °C) must reject, retry once with a clarifying question, then fall back to defaults with a flag. Chapter 6 specifies this pipeline.
- **No enthalpy reference state or energy-closure plan.** Every stream needs a single consistent enthalpy basis (elements at 298.15 K recommended) or the energy balance will not close across reactors and the future economics mode is meaningless. The steam system around the reformer (HP steam at ~100 bar raised in the convection bank and secondary waste-heat boiler) is half the energy story of a real plant and deserves an explicit model object even in MVP.

- **Session and warm-start state is undesigned.** The what-if loop’s responsiveness depends on keeping the last converged solution in memory keyed to the session; the plan implies this but never states it. It should be a first-class object (last converged tear vector, parameter set, diff log) so narration can cite before/after numbers.
- **The fault-parameterization surface is undefined.** The v3 operator-training stage needs a whitelist of faultable parameters (fouling factors, approach values, efficiencies, pressure drops) with sane ranges — defining it early costs nothing and keeps the engine’s API stable. Chapter 5 gives the taxonomy.
- **No regression harness.** A golden set of published stream tables (Chapter 9) should become automated tests from the first week: unit-level checks, then train-level, then the loop, then whole-plant mass and energy closure to 10^{-6} relative.

2.4 Layer-by-layer scorecard

Layer	Plan’s position	Grade	Required fix
Deterministic engine	Custom unit ops + sequential-modular solver	A–	Apply C1–C7; add enthalpy basis and Ar species
Intake (LLM → JSON)	Single-shot extraction, cheap model	A	Add strict schema + Pydantic range envelope + retry
Base narration	Script generated once, read by TTS	A–	Sentence-chunked streaming; 60–90 s interruptible beats
What-if loop	Tool-calling agent, warm-started recompute	A	Protect the recompute path (Ch. 8 critical path); optimistic UI
Diagnosis / rationale	LLM grounded in reference material	B–	Underspecified: ground in EFMA/Ullmann chunks; bind claims to engine numbers
3D render	“3D render” in MVP	B	Define budget: draw calls, tag registry, block-out geometry first
TTS layer	Piper / OpenAI / ElevenLabs cost table	B+	Azure 500k free chars/mo or gpt-4o-mini-tts persona; abstraction over providers
Roadmap	MVP → v5 staging	A–	Re-scope: validation is a stage, exports before economics; weeks, not vibes

Table 2.1 — Scorecard by architectural layer. No layer needs replacing; three need specification work.

The pattern across the scorecard is consistent: the plan’s skeleton is strong, and every weakness is a missing specification rather than a wrong idea. That is the best possible pre-build diagnosis — it means the next step is writing details, not rethinking.

3. The Landscape: What Exists, What Is Missing

Before building, it is worth knowing precisely what a chemical engineering student can reach today, because the answer defines the product’s reason to exist. This chapter maps four territories — commercial simulators, free engines, AI-native tools, and training simulators — then states the white space and the threats.

3.1 The cost wall: commercial simulators

The professional tools are excellent and effectively unreachable for an individual student. AspenTech’s academic program licenses universities rather than students; a September 2023 promotion sold students Aspen Plus or HYSYS at \$100 for one product in a limited window (listed value \$2,700), which is the closest any incumbent came to direct student access. Commercial licenses are quote-only and commonly reported between \$10,000 and \$100,000+ per year. AVEVA PRO/II, VMGSim (now Honeywell) and Chemstations’ ChemCAD are quote-only or site-licensed; Honeywell’s UniSim offers 60-day trials. Every one is a Windows desktop install, usually behind a university VPN or VM. None runs in a browser, none narrates, none explains itself.

Product	Access reality for a student	Price signal
Aspen Plus / HYSYS	Via university site license; occasional \$100 promos, one product, limited window	\$10k–\$100k+/yr commercial
AVEVA PRO/II (SimSci)	Quote-only enterprise; no student path	undisclosed
VMGSim (Honeywell)	Quote-only enterprise	undisclosed
Honeywell UniSim	Free 60-day trial, Windows desktop only	trial-ware
ChemCAD (Datacor)	Free only at universities holding site licenses (e.g. UIUC, Alabama webstores)	site license
MaximaLabs	Browser, zero install; free tier for learning, AI copilot included	\$0 / \$299/mo teams

Table 3.1 — The accessibility wall. Note MaximaLabs at the bottom: the only browser-native player, and the closest delivery-model comparable.

3.2 Free engines: the mechanics are already commoditized

DWSIM is the crown jewel of open-source simulation: GPL-licensed, 35+ unit operations, steady-state and dynamic modes, CAPE-OPEN compliance, Python scripting, and active development (DWSIM 10 even ships a patron-preview AI assistant that builds flowsheets from natural-language commands). IDAES-PSE,

the DOE-funded Pyomo framework, is powerful but expert-oriented. COCO, OMChemSim, NeqSim and ThermoPack round out the ecosystem. The honest question this raises is "why build a custom engine at all?" — and it deserves an honest answer.

- **Determinism and narrative control.** Every number the narrator speaks must be explainable at unit-op granularity, in teaching language, on demand. Wrapping DWSIM means wrapping its property packages, convergence behavior and failure modes in someone else's vocabulary.
- **Latency.** An interactive what-if loop wants a warm-started solve in tens of milliseconds; a full .NET flowsheet simulator re-solve is the wrong shape for that, and its recycle convergence failures are opaque to students.
- **Licensing and deployment.** GPL v3 is workable for a free web app but constrains any future commercialization; the pythonnet/.NET server stack is fragile to deploy.
- **The right use for DWSIM is offline:** build the same plant in it once and use it as a validation oracle against the custom engine — a second opinion, not a runtime.

3.3 The AI wave: research is arriving, products are text-only and pro-priced

Academic work is converging on exactly this product's shape: "Text-to-flowsheet" (RSC Digital Discovery, 2026) generates flowsheets from natural language; Sketch2Simulation (AIChE 2026) turns diagrams into HYSYS models; Liang (2026) built an LLM agent for user-friendly process simulation; AutoChemSchematic (2025) closes the loop by verifying DWSIM balances. On the product side, MaximaLabs (browser, 60+ unit ops, AI copilot, free tier, \$299/mo teams) is the closest comparable; Alkali's ProcessMate (YC, 2025) targets professional design and procurement; DWSIM 10's AI assistant is a desktop patron preview. An IChemE webinar in August 2026 pitched an LLM copilot for Aspen Plus in design education. Every single one is text-first, and none is ammonia-pedagogical, voiced, or free-for-students at full capability.

3.4 The training-simulator world: what the pros pay

Industrial operator training simulators pair a high-fidelity dynamic model with an emulated control system, an instructor station that injects malfunctions, and scoring. GSE Solutions has delivered 173 nuclear, 208 fossil and 114 process simulators since 1971; Honeywell, AVEVA, Yokogawa and Emerson sell comparable suites. The price of admission is brutal: refinery-scale OTS installations run \$800,000 to \$3,000,000, in a market estimated (definitions vary wildly) between \$2.4B and \$12.4B and growing near 12% annually. Operator training is an employer-funded experience; no student can buy in. Meanwhile the free educational layer — LearnChemE's 320+ browser simulations from CU Boulder — is excellent but built from single-concept widgets, with no LLM, no full flowsheet, no voice. The only free ammonia interactive found anywhere is Annenberg's "Control a Haber-Bosch Plant," a legacy equilibrium-only exercise. A peer-reviewed Haber-Bosch teaching module (August 2026) proves educator demand — while requiring a commercial ChemCAD license.

3.5 The ammonia demand signal

Ammonia is the hottest molecule in the energy transition: the IEA's Global Hydrogen Review 2025 counts 200+ committed low-emissions projects with ammonia as the dominant hydrogen carrier, and analysts project the green-ammonia market growing from a \$1.5–3.4B base toward \$170B+ by 2035. The workforce signal is concrete: the UK Hydrogen Skills Alliance needs more than 25,000 hydrogen-sector workers by 2030; South Africa's skills audit found 24 required degree programs not yet offered. Students entering this field have no free interactive tool that teaches an actual ammonia train. That is the audience.

3.6 White space and threats

Cross-referencing all four territories, six gaps remain unoccupied. Ranked by defensibility:

- **Voice-native plant simulation.** Every AI copilot found is text-based; nobody speaks the plant aloud or listens for spoken what-ifs.
- **Free full-flowsheet ammonia simulator** (reforming → shift → methanation → compression → synthesis loop → condense → recycle/purge) — verified absent.
- **LLM what-if loop on a deterministic engine, free for students** — MaximaLabs is pro-priced; the research prototypes are not deployed.
- **3D bound to the live solve** — published VR plants are static visualizations; none map real stream data onto equipment as parameters change.
- **Student-accessible fault injection** — OTS-style malfunction training at \$0 instead of \$800k+.
- **Exportable deliverables from a chat session** — a real mass-balance table or PFD a student can submit.

Threat	Overlap	Verdict
DWSIM + DWSIM 10 AI assistant	High on engine; zero on pedagogy, voice, 3D, ammonia	Pre-empt the "why not DWSIM?" objection in-app with a one-screen answer; cite it as validation oracle
MaximaLabs	High on delivery model (browser + AI copilot)	Professional focus, \$299/mo teams; move fast on the student niche before an edu tier appears
LearnChemE (CU Boulder)	Same audience, different format	Partner or channel, not competitor — single-concept widgets complement a whole-plant sim
Alkali ProcessMate (YC 2025)	Medium on "AI copilot" mindshare	Professional procurement focus; watch for a university pivot
AspenTech + academic copilots	Capability king; IChemE 2026 copilot-for-education thread	Access friction persists; keep an eye on hosted academic tiers

Table 3.2 — Threat model. No incumbent currently moves on voice + ammonia pedagogy + free.

Positioning statement. For chemical engineering students and early-career engineers, this is the free, voice-narrated ammonia plant you can interrogate: every number recomputed live, every unit explained out loud, every fault diagnosable — the OTS experience nobody prices for students, wrapped around one plant modeled deeply enough to check against published data.

4. The Hardened Engine: Correlations, Data, Validation

This chapter resolves the plan's open items — stream schema, correlations per unit operation, converter model choice — with implementation-ready numbers. Unless tagged otherwise, values are sourced from the validation library in Chapter 9; the Gillespie–Beattie implementation was additionally re-verified numerically during this research by reproducing the classic equilibrium tables.

4.1 The stream object, finalized

Field	Type	Units (internal)	Notes
id	str	—	Stable tag, e.g. S-104 (feeds the PFD/3D tag registry)
phase	enum	—	vapor / liquid / two-phase marker for the condenser
T	float	K	Kelvin internally; display °C
P	float	Pa	Pascals internally; display bar
F	float	kmol/h	Molar basis keeps stoichiometry linear
y	dict	mole fractions	9 species; $\Sigma = 1.0$ enforced at 10^{-10}

Table 4.1 — Stream schema. Species set: H_2 , N_2 , NH_3 , CH_4 , H_2O , CO , CO_2 , O_2 , Ar.

Design rules that prevent whole classes of bugs: store mole fractions only (derive mass flow on demand — never two sources of truth); fix one enthalpy reference state for the entire flowsheet (elements at 298.15 K) so reactor duties close; carry units metadata on every object; keep unit-op derived state (duties, approaches, efficiencies) in unit objects, not streams. Tolerances: compositions to 10^{-10} , flows to 10^{-6} relative.

4.2 Unit-operation modeling table

The complete first-build recipe, one row per unit operation. "Eq." gives the governing relation; defaults are the values to hard-code before the first what-if ever runs.

Unit	Model	Key relations and defaults
Feed desulfurization	Black box	Pass-through with ZnO polish to below 0.1 ppm S; narrative hook only
Primary reformer	SR1 + WGS simultaneous equilibrium, ideal-gas K_p , Dalton's law	$K_{SR1} = 1.198 \times 10^{13} e^{-26830/T}$ [bar ²]; $K_{WGS} = 1.767 \times 10^{-2} e^{4400/T}$; 25–35 bar; S/C = 3.0; outlet 780–830 °C; CH ₄ approach 5–10 °C; only 30–40% of hydrocarbon converts here
Secondary reformer	Two-zone: adiabatic air combustion, then equilibrium	Combustion of O ₂ (burns H ₂ first) to ~950–1000 °C; then SR1+WGS to 0.2–0.3% dry CH ₄ ; air flow = DOF setting H ₂ /N ₂ = 3; Ar = 0.93% of air
HT shift	WGS equilibrium + approach	Fe–Cr, inlet ~350–400 °C; exit ~3% CO dry; approach 10–25 °C
LT shift	WGS equilibrium + approach	Cu–Zn, 200–220 °C; exit 0.2–0.4% CO dry; approach 15–30 °C
CO ₂ removal	Splitter + duty	Removal 99.5%+; residual 100–1000 ppmv CO ₂ ; reboiler 30–60 MJ per kmol CO ₂
Methanator	Adiabatic equilibrium	Ni catalyst, ~300 °C inlet; $\Delta T = 74$ °C per 1 vol% CO, 60 °C per 1 vol% CO ₂ ; exit below 1 ppm carbon oxides
Syngas compressor	Closed-form polytropic, 3–4 stages, intercooled to ~40 °C	$(n-1)/n = (k-1)/(k \cdot \eta_p)$; $H_p = Z \cdot R \cdot T_1 \cdot (n/(n-1)) \cdot [(P_2/P_1)^{(n-1)/n} - 1]$; $\eta_p = 0.75$; $k \approx 1.40$ (3:1 H ₂ /N ₂); $Z \approx 1.02$ –1.08 (H ₂ -rich, do not assume below 1); 25–35 bar → 150–200 bar
Synthesis converter	3 adiabatic beds; per-bed equilibrium + 0.90–0.95 fractional approach	Bed inlets 380–430 °C; outlets capped 500–530 °C; $\Delta H = -46$ kJ/mol NH ₃ drives the adiabatic rise; outlet 14–18 mol% NH ₃ ; inlet (recycle) ~2 mol%
Condenser / separator	PR-EOS flash (thermo library) with BIPs	Chill to ~-20 °C (2-stage NH ₃ refrigeration, COP $\approx$ 2.4); gas leaves with ~2–3% NH ₃ ; liquid product 99.5 wt%
Circulator	Small booster compressor	ΔP 5–25 bar; $\eta \approx 0.75$
Splitter (recycle/purge)	Pure mass balance	Purge 5–8% of loop gas, holding inerts (CH ₄ +Ar) at 10–15 mol%; recycle multiple $\approx$ 4–5× makeup

Table 4.2 — First-build modeling recipe. Sources: EFMA (2000); Flórez-Orrego and de Oliveira Junior (2017); Zečević and Bolf (2020); Johnson Matthey methanation data; Rice CENG403.

Two structural notes on the table. First, the BAT loop arrangement adds makeup gas *after* ammonia condensation and takes purge after the separator and before makeup — getting this order right is the difference between matching published loop compositions and not. Second, the equilibrium-versus-kinetics question resolves cleanly: equilibrium with fractional approach reproduces plant outlets within 1–2 points of NH₃ with zero catalyst data, and the full Dyson–Simon Temkin–Pyzhev kinetics (constants in Flórez-Orrego; refits in the 2024 literature) become worth adding only when the product needs bed temperature profiles, catalyst loading, or part-load behavior. Keep kinetics out of the first build; design the converter API so they can drop in later.

4.3 The ammonia equilibrium constant, implementation-ready

For $\frac{1}{2}\text{N}_2 + (3/2)\text{H}_2 \rightleftharpoons \text{NH}_3$, the Gillespie–Beattie correlation in partial-pressure (atm) terms:

$$\log_{10} K_p = 2074.8/T - 2.4943 \cdot \log_{10} T - \beta \cdot T + 1.856 \times 10^{-7} \cdot T^2 + I$$
$$K_p = p_{\text{NH}_3} / (p_{\text{N}_2}^{0.5} \cdot p_{\text{H}_2}^{1.5}) \text{ [atm}^{-1}\text{]}$$

with β and I interpolated linearly in pressure between the rows below. Valid from 325–950 °C and 10–300 atm — do not extrapolate beyond 300 bar. Beware the convention trap: if the reaction is written $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, the equilibrium constant is the square of the half-reaction’s.

P (atm)	β	I
10	0	1.993
30	3.4×10^{-5}	2.021
50	1.256×10^{-4}	2.090
100	1.256×10^{-4}	2.113
300	1.256×10^{-4}	2.206

Table 4.3 — Pressure-dependent coefficients (Gillespie and Beattie, Phys. Rev. 36, 1930; table verified via El-Gharbawy 2021).

The implementation was re-verified during this research by solving the equilibrium composition for a stoichiometric feed at a grid of conditions; the resulting equilibrium NH_3 fractions reproduce the classic Larson–Dodge/Vancini tables exactly, which is the strongest cheap correctness check available for this correlation.

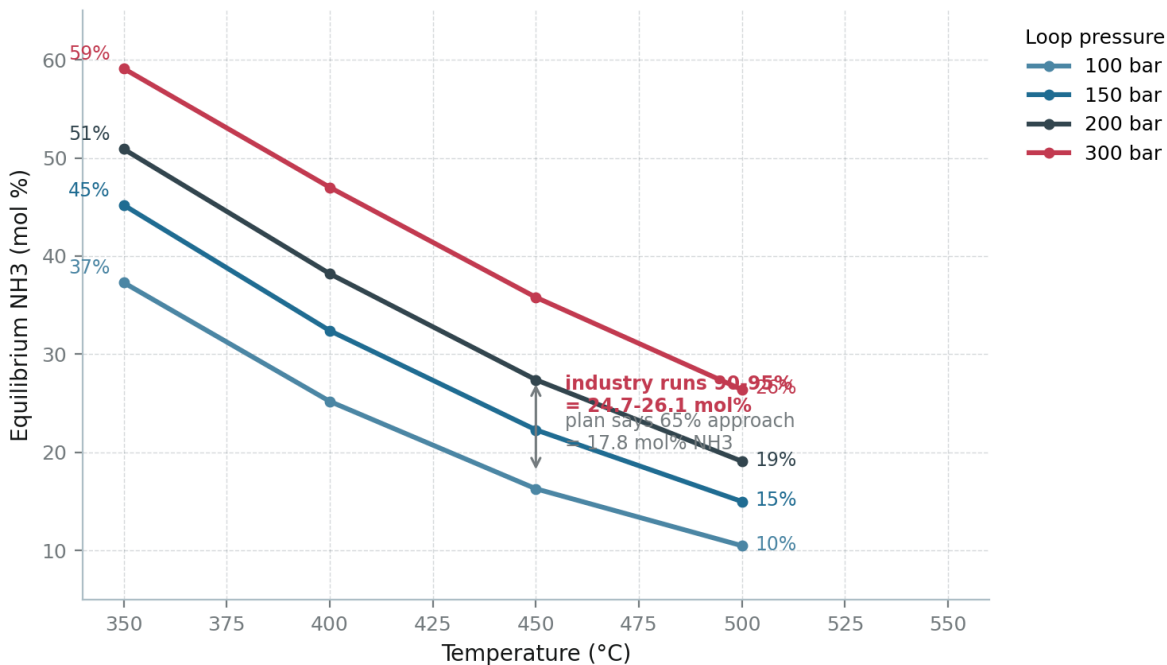


Figure 4.1 — Verified equilibrium NH_3 fraction versus temperature at four loop pressures (computed from the Table 4.3 correlation; spot-matches the classic tables). The annotated gap at 450 °C / 200 bar shows correction C1: the plan’s 65% approach predicts 17.8 mol% where industry operates at 24.7–26.1 mol% (90–95% approach).

4.4 Convergence strategy for the synthesis loop

Tear the recycle at the converter inlet (the standard choice) and solve the loop with a staged strategy rather than a single method:

1. **Iterations 1–3:** direct substitution with 0.5 damping — robust, gets into the correct basin while inert levels settle.
2. **Iterations 4–25:** bounded Wegstein acceleration per tear variable, with the acceleration factor clipped to $[-5, 0]$ exactly as commercial simulators do; unbounded Wegstein is the classic cause of loop divergence.
3. **Stall fallback:** if not converged in ~25 iterations, switch to Broyden quasi-Newton on the 6–8 tear variables (component flows of N_2 , H_2 , NH_3 , CH_4 , Ar plus total flow and temperature). As a last resort, Luyben-style dynamic relaxation integrates the loop to steady state.
4. **Always:** initialize the tear stream near the answer — 13 mol% inerts, 2.5 mol% NH_3 , $H_2/N_2 = 2.7$, 380 °C, loop pressure. A cold start from pure 3:1 syngas wastes iterations and invites the trivial no-reaction root.

Expected cost: 10–20 loop evaluations cold; with warm-starting, a small what-if perturbation typically converges in a handful of evaluations — the entire basis for a conversation-speed recompute. Tolerance: 10^{-4} to 10^{-5} relative on tear molar flows.

4.5 Validation plan: what "correct" means, numerically

Internal consistency is not validation. The engine earns credibility against four external anchors (Chapter 9), applied in escalating order: unit-level spot checks, then the once-through front-end train, then the loop, then whole-plant closure. The quantitative gates:

Check	Target	Anchor
Whole-plant mass balance closure	below 10^{-6} relative	internal
Energy balance closure	within $\pm 2\%$	internal
Major stream compositions	within $\pm 1\text{--}2$ mol% absolute	EFMA ranges; TAMU stream tables
Front-end temperatures and CH_4 slip	within ± 15 °C / $\pm 1\%$ dry CH_4	EFMA; Rice CENG403
Converter outlet NH_3	within 10% relative of 14–18 mol%	Flórez-Orrego; Shamiri
Loop inerts and purge	inerts 10–15 mol%; purge 5–8%	EFMA; Flórez-Orrego
Specific energy consumption	28–31 GJ/t NH_3 (BAT band)	IFA/Frisse; IEA

Table 4.4 — Validation gates. The golden stream tables become permanent regression tests.

4.6 Energy numbers worth hard-coding

Best-available-technology SMR plants consume 28–29 GJ per tonne of NH_3 (Uhde's fleet: 6.6–7.2 Gcal/t); the 2008 IFA survey averaged 36.6 GJ/t across 93 plants and the IEA's global average sits near 41 GJ/t. These bands are what the v5 economics mode should reproduce before it earns opinions. For contrast, green ammonia needs 10–12 MWh of electricity per tonne (electrolysis at ~ 53 kWh/kg H_2 plus air separation, compression and the loop), which at \$30–70/MWh yields \$300–840/t of energy cost against \$80–270/t for natural gas at \$3–8/MMBtu — a teaching-ready trade-off. Within the plant, the syngas compressor dominates the loop's power draw at 61–75% (a 1000 MTPD plant totals ~ 13 MW optimal), and the reformer furnace is the largest fuel consumer (~ 58 MW at 1000 MTPD in the Rice design); HP steam at ~ 100 bar raised across the convection bank, secondary waste-heat boiler and synloop boilers drives the compressor turbines. Modeling even a simplified steam header makes the energy narrative honest.

5. The Interactivity Frontier

The plan's roadmap already gestures at an "extremely interactive" product; this chapter maps how far that can actually go in 2025–2026, what each step costs a solo developer, and which patterns make the difference between a demo that impresses for a minute and one that people refuse to close. Three frontiers are explored in depth — live voice, 3D space, and gamified operations — followed by the design science that should govern all of them.

5.1 The interactivity ladder

Fifteen concrete features, ranked by wow-per-unit-effort. Effort figures assume a competent solo developer already holding a working engine; they exclude the engine itself.

#	Feature	Effort	Depends on	Risk
1	3D stream-flow shader — UV-scroll on pipe meshes; speed $\propto$ mass flow, color $\propto$ temperature	2–3 d	Three.js tube geometry + engine rates	Low
2	Live PFD — React Flow with marching-dash streams and live T/P/flow labels	2–4 d	React Flow (MIT); memoized nodes	Low
3	Sentence-chunked streaming narration — TTS starts on the first sentence	1–3 d	Streaming LLM + streaming TTS	Low
4	Interruptible narration — 60–90 s beats, pause/resume/discard on any input	2–5 d	Beat timeline; browser audio	Low–Med
5	Exploded-view slider + click-to-inspect unit cards	3–5 d	R3F; tag registry	Low
6	Push-to-talk what-if — STT $\rightarrow$ LLM tool-call $\rightarrow$ recompute $\rightarrow$ spoken answer	3–7 d	STT; tool-calling LLM	Medium

#	Feature	Effort	Depends on	Risk
7	Fault-injection operator mode (steady-state) — fouling, deactivation, efficiency, dP	1–2 wk	Fault whitelist; alarms; KPI deltas	Medium
8	P&ID; ↔ 3D bi-directional sync — no web precedent found	1–2 wk	Shared tag registry; camera tweens	Medium
9	Scenario challenges + scoring — briefing → ops → score → debrief	1–2 wk	Feature 7; scoring engine	Medium
10	First-person walkthrough with proximity narration	1–2 wk	R3F controls; LOD	Medium
11	Full-duplex "talk to the plant" — Realtime API with barge-in	1–3 wk	OpenAI Realtime or LiveKit/Pipecat	High
12	Instructor station — set initial conditions, inject faults remotely	2–4 wk	Features 7 + 9	Medium
13	Quasi-dynamic plant clock — steady solve per game tick + inventory integrators	1–3 wk	Solve time under 0.5 s; holdup data	High
14	AI-authored fault briefs — LLM writes scenario + rubric, engine validates	3–7 d	Fault whitelist guardrails	Medium
15	Leaderboards + shareable scenario cards	3–5 d	Feature 9; light backend	Low

Table 5.1 — The interactivity ladder. Rungs 1–6 form the "alive plant" tier; 7–9 the operator-training tier; 11+ the headline-demo tier.

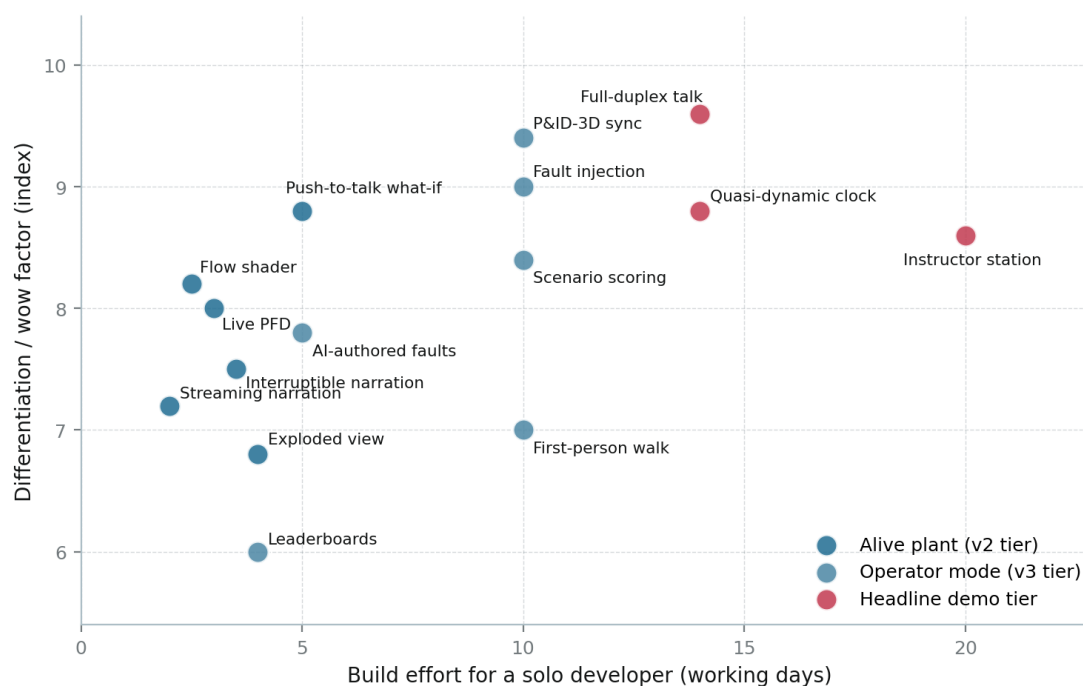


Figure 5.1 — The ladder in the effort-impact plane. Color encodes roadmap tier: blue = alive plant (stage S4), teal = operator mode (S5), red = headline-demo tier. The upper-left cluster is the highest-return work available to this product.

5.2 Frontier A — live conversational voice

The perceived-quality budget for voice is brutally concrete. Nielsen's thresholds still govern: 0.1 s feels instantaneous, 1 s is the limit of flow, 10 s is the attention limit. Voice specifically: humans take turns with a 200–300 ms gap; first audio under 600 ms feels natural and under 400 ms feels excellent — while the median production voice agent today still answers in 1.4–1.7 s. The good news is that the gap is closable with engineering, not research.

Interaction	Budget	Technique
Slider drag / camera move	under 0.1 s	Client-side interpolation; precomputed solution surfaces
What-if recompute + visual update	0.5–2 s	Warm-started solve; optimistic UI — animate flows toward new rates immediately
Narration start after an event	under 0.6 s	Sentence-chunked streaming TTS (first ~24-token segment)
Voice Q&A; turn	under 1 s (ideally 0.6)	Realtime API, or Deepgram STT + LLM + Flash-class TTS pipeline
Fault scenario tick	0.5–2 s / step	Quasi-steady solver stepping
Anything slower than 1 s	visible feedback	Pulsing pipes, status chip, or a spoken filler ("let me re-solve the loop")

Table 5.2 — Latency budget for the product. Every row has a known technique; none requires research.

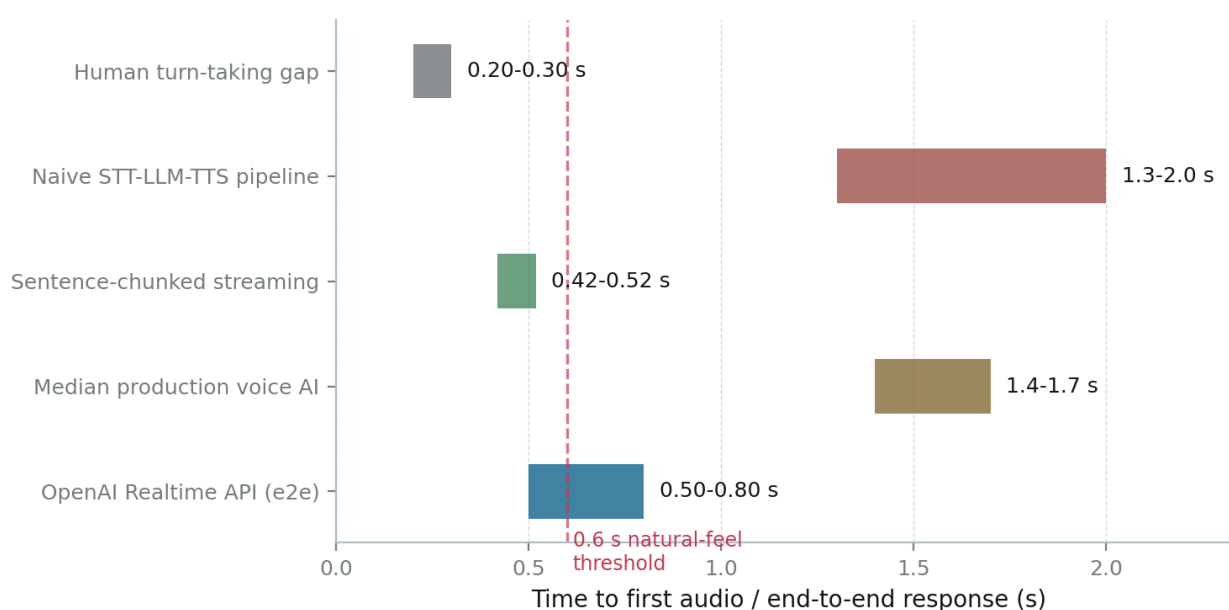


Figure 5.2 — Achievable voice latencies by architecture, against the ~0.6 s natural-feel threshold. Sentence-chunked turn-taking already lands inside the band; full-duplex speech-to-speech closes most of the rest.

Two architectures, staged. The turn-based pipeline (STT → LLM → TTS) naively costs 1.3–2 s, but sentence-chunked streaming — start TTS on the first sentence while the LLM keeps generating — reaches 0.42–0.52 s and needs only commodity APIs. That is the right v2 path. Full-duplex speech-to-speech (OpenAI Realtime over WebRTC, with function calling so the model can invoke the engine

mid-conversation) delivers 0.5–0.8 s with true barge-in at roughly \$0.04/min of speech after the 2025 price cuts — the right v3+ "live call" mode. Self-hosted alternatives (LiveKit Agents, Pipecat) exist if API cost or data residency demands them. The browser's built-in Web Speech API is the free fallback tier — good enough for a mute/read mode and always-available on desktop Chrome and Safari, but unreliable as the flagship (no custom vocabulary: it will mangle "Wegstein" and "Haber").

Interruptible narration is a design pattern, not a feature. Museum audio-guide craft suggests the shape: segment the walkthrough into 60–90 s stops (110–170 spoken words), each tied to a visual event — a unit assembling, a stream lighting up. Any click, key press or speech pauses audio; if the user resumes quickly, play on; if they keep talking, discard the pending beats and answer. This makes narration seekable like a timeline, which also solves the "replay that part about the purge" request every teacher will make. Be warned that interruption handling in production voice AI is still genuinely hard — barge-in triggers on filler words and echo — so the pause gesture should be explicit (tap or spacebar) in v2, with true open-microphone barge-in deferred to the Realtime-API stage.

5.3 Frontier B — the 3D spatial plant

The performance currency of the web is draw calls, not triangles: scenes under ~100 draw calls run at 60 fps on most laptops, above ~500 even strong machines struggle. An ammonia plant — a dozen equipment groups plus a pipe rack — fits comfortably in 50–300 draw calls and 100–500k triangles once geometry is merged and instanced, which means 60 fps on the integrated GPUs students actually own. Three.js with React Three Fiber is the clear stack choice: it loads in 2–6 s where Unity WebGL takes 8–30 s, runs at 95–99% of raw WebGL performance, and — decisively for this product — lives inside the same React state tree as the engine results. Unity would fight the stack for no gain.

- **Flow animation:** scroll a stripe texture along each pipe's UVs (speed proportional to the solved mass flow, color to temperature) — roughly one draw call per material for every stream in the plant. Reserve GPU particles for two or three hero streams (feed gas, liquid ammonia product); real fluid solvers are overkill.
- **The 2D twin:** a live PFD in React Flow (MIT-licensed, nodes are React components, so live-updating stream labels are trivial) with marching-dash animation on stream edges. Perf hygiene matters — memoize nodes or re-renders will crawl — but the pattern is proven at 300+ nodes, and Siemens ships a React Flow modeler for process diagrams as validation of the approach.
- **The differentiator:** bi-directional P&ID;—3D sync. Click a vessel in 3D, it highlights on the PFD with a camera fly-to; click a stream on the PFD, the pipe glows in 3D. Research found this only in enterprise plant-design suites (AVEVA), never on the web. Implementation is unglamorous plumbing — a shared equipment-tag registry mapping tag → PFD node id → 3D object id → engine unit id, plus selection events and camera tweens — which is exactly why it is defensible.
- **Day-one restraint:** the MVP needs block-out geometry (procedural cylinders, boxes and tube runs) with correct spatial arrangement, not photorealism. Exploded-view sliders and click-to-inspect cards (rung 5) deliver more perceived quality per hour than texture work ever will.

5.4 Frontier C — gamified operations

The OTS industry's formula is dynamic model + emulated controls + instructor station + scoring, at \$800k–\$3M a seat. A steady-state engine cannot honestly replicate all of that — and pretending otherwise would be the fastest way to lose an engineer's respect. The honest version is "sensitivity analysis dramatized," and it is genuinely pedagogical:

Fault (steady-state-honest)	Parameter delta	What the student sees
Condenser fouling	lower U / larger approach	Colder separator temperature needed; NH ₃ losses up; refrigeration duty up
Catalyst deactivation	bed approach 0.95 → 0.85	Outlet NH ₃ falls; recycle and purge flows rise; production drops
Compressor degradation	η_p 0.75 → 0.68	Power draw climbs; energy KPI degrades; steam balance tightens
Feed drift	H ₂ /N ₂ off 3:1	Conversion collapse the student must diagnose and correct
Inert buildup	purge partially blocked	Loop pressure and inerts creep; the purge story becomes physical
Cooling-water degradation	CW temperature +10 °C	Condensation worsens; loop efficiency falls
Pressure-drop fault	dP multiplier ×1.5	Circulator work up; a lesson in hydraulic design margin

Table 5.3 — Steady-state fault taxonomy: every fault is a parameter delta → re-solve → KPI deltas and alarms. This is the v3 operator mode.

What steady state cannot do honestly: trips, startup and shutdown sequences, compressor surge, relief events, oscillations, and anything scored on response time. The bridge is **quasi-steady stepping** — re-solve the plant every 0.5–2 s of "plant time," interpolate between solutions, and treat drum and loop inventories as simple integrators so levels drift as faults develop. Many educational simulators do exactly this; the requirement is honest labeling ("quasi-steady") and a solver fast enough for the tick, which the warm-started engine satisfies. Scoring, borrowed from OTS practice where the debrief is where learning happens: keep-operating-band, time-to-diagnose (once the clock exists), final production and energy, alarms acknowledged — wrapped in a briefing → operations → score → debrief loop. Evidence from the serious-games literature (2023–2025 systematic reviews) supports the mechanics: points, challenges and scenarios reliably lift engagement and help retention, with the strongest learning gains coming from scenario-plus-debrief structure rather than decoration. GSE's own 2026 move — AI authoring fault scenarios from natural language — is the industrial blessing of exactly this feature (rung 14).

5.5 The design science underneath

PhET's Colorado sims, the most studied interactive learning artifacts in existence, distill to five principles that map onto this product almost verbatim: direct manipulation with immediate feedback (grab the parameter, see the plant answer); make the invisible visible (flow speed as animation speed, temperature as color); real-world connections (this is a real plant with real published numbers); minimal text (the voice carries the explanation); constrained controls that scaffold exploration (sliders bounded by physical ranges,

not free text fields). The explorable-explanations tradition adds two more: one-step interactions — every drag should produce a visible consequence — and persistent context: never collapse the old state when showing the new, which for this product means the what-if answer must show before/after side by side, keeping the prior plant visible for comparison. These are cheap rules to follow at design time and expensive to retrofit; they belong in the first wireframe, not the second release.

6. Recommended Technology Stack

The plan's candidate list (Cantera, CoolProp, Piper, OpenAI, ElevenLabs, Azure) was verified against current pricing, licensing and capability. Verdicts below are for this product specifically — a teaching app with a deterministic engine, voice narration and a student audience — not general rankings.

6.1 The stack, layer by layer

Layer	Primary pick	Budget alternative	Why
Engine core	Custom unit ops + hand-coded K_p correlations	Same (there is no shortcut)	Determinism, narrative control, warm-start latency
Loop VLE + enthalpies	<i>thermo</i> (MIT): PR-EOS with BIPs, flashes	CoolProp pure-fluids + K-value condenser shortcut	Only mainstream library that flashes NH_3 mixtures at loop conditions
Pure-fluid properties + steam	CoolProp (MIT; NH_3 EOS to 725 K / 1000 MPa; IF97 steam)	<i>iapws</i> for steam only	Bulletproof pure-fluid layer; never use it for the mixture flash
Equilibrium cross-check	Cantera (BSD): Gibbs minimization, offline	Skip	Independent second opinion on reformer/shift equilibrium
Intake + what-if LLM	OpenAI structured outputs (strict schema) via Instructor + Pydantic	Any provider through the same Instructor layer	100% schema adherence; reliability lives in the Pydantic envelope, not the vendor
Narration TTS	OpenAI gpt-4o-mini-tts (~\$0.015/min; voice-instructions persona)	Kokoro-82M self-host (Apache 2.0) or Azure free tier	Persona control ("calm plant instructor") is a pedagogical feature
Conversational TTS	ElevenLabs Flash v2.5 (75 ms TTFB, \$50/1M chars)	Azure Speech free tier (500k neural chars/month)	Latency class for the Q&A; loop
STT	Deepgram Whisper-class (~150 ms)	Browser Web Speech API	Accuracy-speed balance; word boosting for jargon
Full-duplex voice (v3+)	OpenAI Realtime API over WebRTC (~\$0.04/min)	LiveKit Agents or Pipecat self-host	Barge-in conversations that call the engine mid-turn
3D render	Three.js + React Three Fiber	Babylon.js	2–6 s loads; lives in React state; 60 fps within draw-call budget
2D PFD	React Flow (MIT)	AntV X6	Nodes are React components → live labels are trivial
Validation oracle	DWSIM (offline, scripted via Python)	None	Second opinion on the whole flowsheet; GPL is fine offline

Table 6.1 — Recommended stack. All primary picks are free or metered-in-cents for this scale.

6.2 TTS economics, verified

TTS pricing moved decisively in favor of small builders: Amazon gutted its perpetual free tier in July 2025 (replaced by one-time \$200 credits — skip Polly), while Azure’s 500k free neural characters per month remains ongoing, Google gives 4M standard characters free monthly, and OpenAI’s mini TTS model prices at roughly \$0.015 per minute with a voice-instructions channel that can lock an instructor persona. At the top end, ElevenLabs remains the quality ceiling at \$50–100 per million characters with a 75 ms time-to-first-byte Flash tier built for agents.

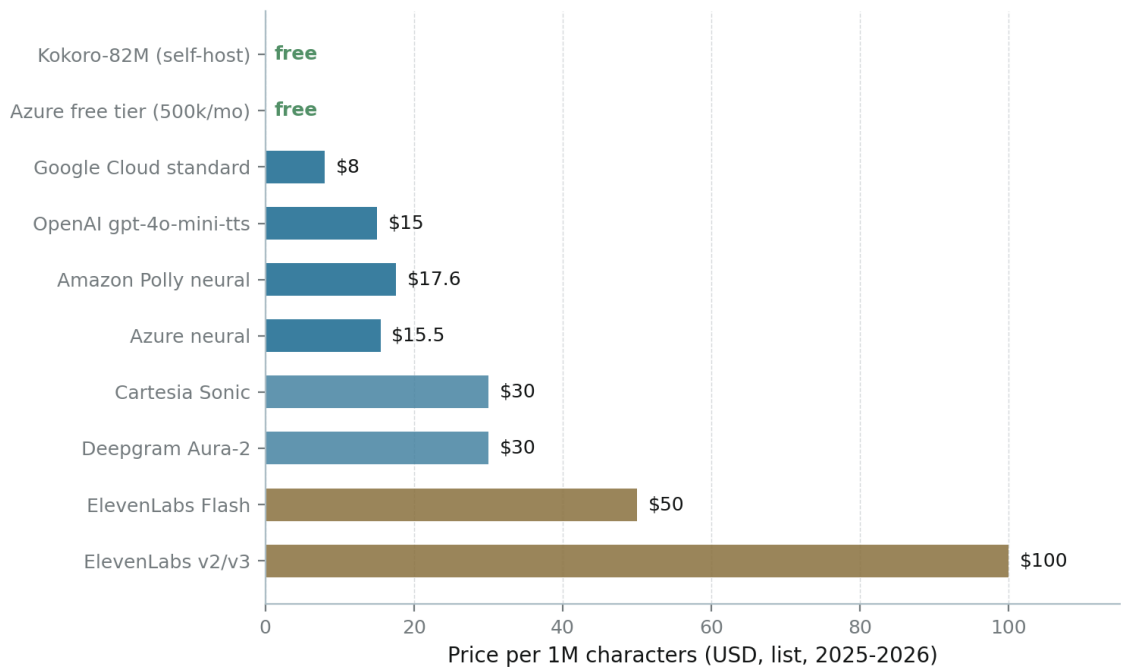


Figure 6.1 — TTS list prices per million characters (2025–2026). Green bars carry a genuine free tier; the vertical spread is 100×, yet a semester of narration costs pennies even at the top.

Scenario	Compute	Cost
100 pre-generated walkthroughs × 3 min	OpenAI mini-TTS	≈ \$4.50 — or \$0 within Azure’s 500k free monthly chars
500 conversational answers × 30 s	ElevenLabs Flash	≈ \$12.50 — or \$0 (Kokoro self-host / Azure free)
A "live call" session, 20 min	OpenAI Realtime	≈ \$0.80 — plus ~\$0.20/h idle keep-alive

Table 6.2 — TTS cost sanity check. Voice quality and latency are the real decisions; money is not.

6.3 The LLM integration pattern, specified

The reliability stack has four stages, and the interesting truth is that the vendor provides only the first: schema guarantees are format-level, not value-level — a model can return 900 bar in perfectly valid JSON. Numeric hallucination is a documented LLM failure class with engineering-specific propagation risks, so the architecture must make bad values die before they reach the engine.

1. LLM call with strict JSON schema (enums for units, additionalProperties: false)

2. Pydantic validation with hard physical ranges:
loop_pressure: 80–250 bar S/C ratio: 2–4 bed temps: 350–530 °C
converter approach: 0.85–0.97 purge fraction: 0.02–0.15

3. On rejection → one clarifying retry → then defaults + visible flag

4. Engine computes everything numeric; the LLM only narrates engine output,
with numbers injected server-side into its templates

Instructor (MIT) provides the retry plumbing provider-agnostically, so OpenAI, Anthropic or a local model can sit behind the same interface — worth building that abstraction on day one not for ideology but because pricing and capability will keep moving. The fallback ladder for open-ended questions ("why does the purge stream exist?") is retrieval over a small corpus of grounded material — the EFMA booklet, Ullmann's ammonia chapter, the sources in Chapter 9 — with the engine's live numbers interpolated into the answer. Every physical claim the product makes should be traceable either to a tool call or to a retrieved passage; that traceability is the product's integrity, and it is also the demo you show skeptics.

6.4 Cost model at three scales

Scale	Assumptions	Monthly cost, paid stack	Free-tier path
Single user (you)	40 sessions; 10 what-ifs each	≈ \$3–6	≈ \$0
One classroom (30 students)	3 sessions/student/month; voice on	≈ \$25–60	≈ \$0–5
Public launch (1,000 users/mo)	2 sessions/user; 20% voice	\$250–900	\$30–80 (Kokoro + Azure free + rate limits)

Table 6.3 — Operating cost at three scales, dominated by LLM tokens and realtime voice minutes. The free-tier path exists at every scale.

The strategic reading of this table: cost never blocks the mission. Even the paid stack at public-launch scale is a rounding error against any institutional budget, and the free-tier path (self-hosted Kokoro, Azure's perpetual free characters, Web Speech fallback, aggressive caching of narration for repeated parameter sets) keeps the product sustainable at zero. The costs that actually matter are the ones money cannot fix: solver latency, voice naturalness on jargon, and the honesty of the what-if loop.

7. Risk Register

Risks are ranked by expected damage to the product's core promise: an engineer-checkable plant behind an extremely interactive face. Impact (I) and likelihood (L) are scored high/medium/low; every risk carries a mitigation that is already implied by earlier chapters.

ID	Risk	I	L	Mitigation
R1	Recycle convergence stalls or diverges on what-if edge cases, breaking the "instant recompute" feel	H	M	Bounded Wegstein + Broyden fallback (4.4); clamp what-if parameter deltas; on failure, fall back to last converged state with an honest message
R2	LLM emits schema-valid but physically absurd parameters	H	M	Pydantic physical-range envelope + one clarifying retry + defaults with visible flag (6.3)

ID	Risk	I	L	Mitigation
R3	Narration drifts from the engine numbers under iteration pressure	H	L	Templates inject numbers server-side; LLM frames but never computes; regression test compares spoken text to engine output
R4	PR-EOS condenser split deviates from plant data beyond teaching tolerance	M	L	Validate flash against Flórez-Orrego loop tables; fall back to K-value correlation; cite Herrmann 2020 as reference target
R5	Free-tier policy changes (Polly precedent, July 2025) strand the voice budget	M	M	Provider abstraction layer; Kokoro self-host as the permanent floor; cache narration audio by parameter hash
R6	Voice latency misses the 0.6 s bar and the product feels sluggish	H	M	Sentence-chunked streaming as the default; precompute first-sentence audio for canonical questions; optimistic visuals while solving
R7	Scope creep beyond ammonia (CO ₂ plants, methanol, "just one more unit op")	H	H	The roadmap's stage gates; new plants are a v6+ question, and the architecture already isolates unit ops for that day
R8	3D scene drops frames on school laptops	M	M	Draw-call budget (≤ 300) enforced in code review; instancing/merging; LOD; measure on integrated GPUs, not the dev machine
R9	DWSIM 10's free AI assistant neutralizes the "why build this?" story	M	M	Differentiate on pedagogy, voice, ammonia depth and the student experience; keep the one-screen "why not DWSIM" answer in-product
R10	MaximaLabs or an incumbent ships a free edu tier first	M	M	Speed: the niche is small enough to be unattractive to them for a while; the ammonia-specific narrative is hard to retrofit
R11	Solo-maintainer burnout; the 27-week plan slips	H	H	Stage gates sized for sustainable part-time cadence; the MVP demo lands at week 13 precisely to pay motivation forward
R12	API cost blowout if a demo goes viral	M	M	Free-tier-first stack (6.4); per-session rate limits; narration caching; realtime voice behind an opt-in toggle
R13	ASR mangles process jargon ("Wegstein", "Haber", "Temkin")	M	M	Word boosting / custom vocabulary on Deepgram; push-to-talk UX; show the recognized text for correction before solving
R14	Pressure to chase green-ammonia scope before the SMR core is validated	M	L	Electrolysis front-end is a parameter set + one new unit in v5+; do not touch before S2 validation gates pass
R15	Teaching content carries an unverified number into the world	M	M	Every number in narration traces to Chapter 9 sources or engine output; rationale mode cites passages, not memory

Table 7.1 — Risk register, ranked by expected damage. R7 and R11 are the two high-likelihood items, and both are roadmap discipline problems rather than technical ones.

Read as a set, the register says something useful: the technical risks are all mitigated by specifications already in this dossier, and the two risks most likely to kill the project are behavioral — scope creep and burnout. The roadmap in the next chapter is designed around exactly those two.

8. Roadmap v2: Stages, Effort and the Critical Path

The original five-stage roadmap (MVP → what-if → faults → exports → economics) keeps its shape but gets three changes: validation becomes a stage with gates instead of a sentence; the MVP's 3D render is re-scoped to block-out geometry so the demo lands at week 13 instead of week 20; and every stage carries an effort estimate in working weeks for a solo developer working part-time (roughly halve them full-time).

8.1 The stages

Stage	Weeks	Deliverables	Exit gate
S0 — Foundations	0–2	Stream object (Table 4.1); thermo core (<i>thermo</i> + CoolProp); splitter/mixer; plumbing tests; regression harness skeleton	Mass balance closes to 10^{-6} on synthetic networks
S1 — Front-end train	2–5	Desulfurizer, both reformers (two-zone secondary), HTS/LTS, CO ₂ removal, methanator, compressor	Front-end matches EFMA/Rice ranges (Table 4.4 rows 3–4)
S2 — Synthesis loop + validation	5–9	Converter (3 beds, 0.90–0.95 approach), PR-EOS condenser, circulator, splitter; tear solver (4.4); DWSIM cross-check	All Table 4.4 gates pass; golden stream tables committed as tests
S3 — MVP shell	9–13	Intake LLM + Pydantic envelope; narration script from real results; TTS; block-out 3D scene; guided voice walkthrough	A chemical engineer builds a plant from a spec and hears true numbers
S4 — v2 alive plant	13–17	What-if tool-calling loop with warm start; live PFD (rung 2); stream-flow shader (rung 1); streaming + interruptible narration (rungs 3–4); push-to-talk (rung 6)	Ten canonical what-ifs answered in under 2 s with recomputed numbers
S5 — v3 operator mode	17–22	Fault whitelist + alarms + KPI deltas (rung 7); scenario scoring + debrief (rung 9); AI-authored briefs (rung 14)	A full fault scenario plays start to debrief
S6 — v4 exports	22–24	Mass/energy balance tables; PFD image export; session report	Exported tables pass the same validation gates as the engine
S7 — v5 economics	24–27	Energy integration numbers, SMR vs green-ammonia trade-offs, design comparison mode	BAT band 28–31 GJ/t reproduced; comparison mode demos cleanly

Table 8.1 — Roadmap v2. Headline-demo tiers beyond S5 (full-duplex talk, instructor station, quasi-dynamic clock) are deliberately unestimated until S5 ships.

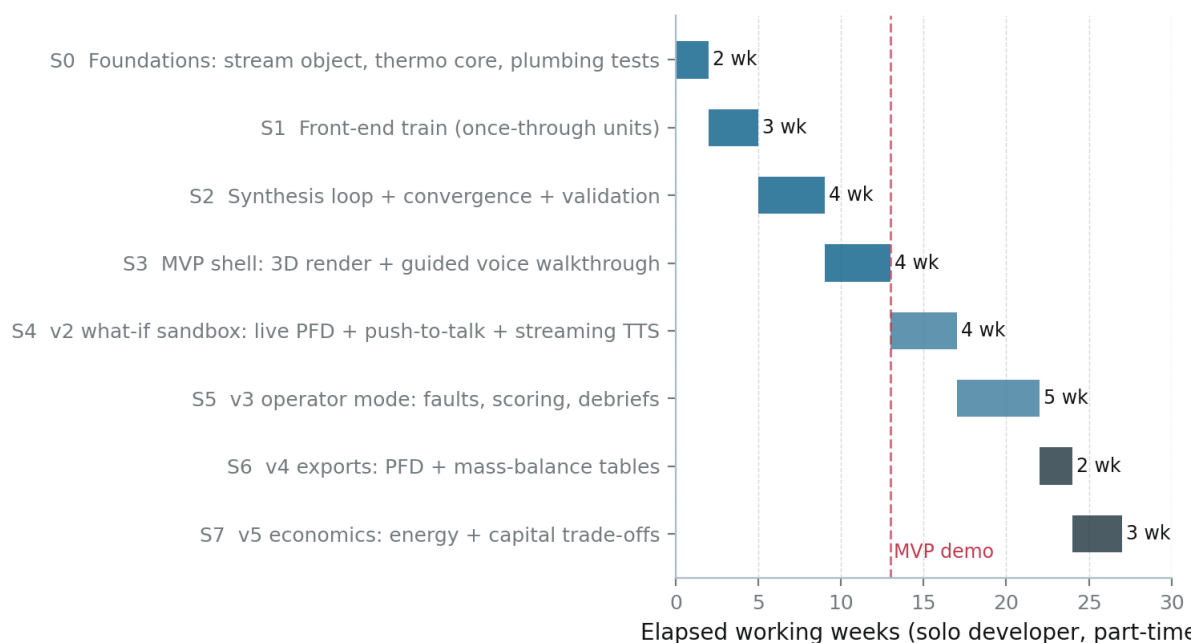


Figure 8.1 — Roadmap v2 on the calendar. The MVP demo at week 13 is the motivation engine for everything after it; validation (S2) is the credibility engine.

8.2 What changes from the original plan, and why

- **Validation is stage S2, not an afterthought.** The original plan's build order ends with "validate against a published mass/energy balance" — correct instinct, wrong position. The gates in Table 4.4 must pass before the MVP shell consumes engine output, or the narration will memorize wrong numbers and the 3D scene will animate wrong flows.
- **The MVP keeps 3D but drops polish.** The original "3D render" in MVP is the right demo instinct; block-out geometry with the flow shader (2–3 days) delivers 80% of the perceived magic. Photoreal equipment, first-person walking and P&ID; sync move to S4/S5 where they compound with the what-if loop instead of delaying it.
- **Exports moved ahead of economics.** v4-before-v5, unchanged from the plan, and reaffirmed: a student's assignable artifact (the mass-balance table) spreads the product through the channel that matters — coursework — before opinions about money exist.
- **The headline-demo tier is unscheduled on purpose.** Full-duplex conversation, instructor station and the quasi-dynamic clock are the v3+ frontier; estimating them now would be theater. They appear in Table 5.1 with dependencies so the architecture accommodates them cheaply, but the calendar stops at S7.

8.3 The critical path: protect the what-if loop

The original plan contains its own most important sentence — that the gap between MVP and v5 is whether the what-if loop is real or fake — and every finding in this research sharpens it. "Real" has a precise, testable definition: the user's utterance becomes a parameter delta; the delta passes the physical envelope; the engine

recomputes from the warm-started state; the narration reports the before/after diff from engine output. Any path where the LLM describes a consequence without a tool call having produced it is the failure mode. Three defenses keep that path intact:

1. **A truthfulness test in every release gate.** Ten canonical what-ifs with known answers (pressure 250 bar; purge halved; LT-shift approach degraded; and so on) run against every build; a wrong trend or a fabricated number fails the release.
2. **Optimistic UI that never lies.** Flows may animate speculatively toward the new state, but numbers displayed as truth come only from converged solves — label the in-between honestly ("solving...").
3. **Provenance on every claim.** Each spoken sentence about the plant is generated from engine output or a retrieved, citable passage — never free generation. The diagnosis mode inherits this rule, which is what makes it a teacher rather than a chatbot.

The one-sentence version: ship a plant whose numbers are checkable, wrap it in interactions that answer in under a second, and never let the language model near the arithmetic. Everything else in this dossier is detail.

9. Validation Source Library

The ranked anchors for engine validation, followed by the key web sources behind the stack and landscape chapters. For each: what it provides and how to use it. The recommended practice is to transcribe the numeric tables below into a versioned golden stream-table file on day one; the regression harness then compares every build against it automatically.

9.1 Engineering anchors, ranked

#	Source	Provides	Access
1	EFMA / Fertilizers Europe, "Production of Ammonia" BAT Booklet No. 1 (2000)	Typical T/P/composition ranges for every section of a conventional SMR plant — the flowsheet-level sanity check	Free PDF (productstewardship.eu)
2	Flórez-Orrego and de Oliveira Junior, Energy 137 (2017) 234–250 (free ECOS 2016 version)	The most complete public synthesis-loop dataset: 1000 MTPD unit, bed temperatures, per-bed conversions, recycle composition, power split, COP, kinetic constants	Free PDF (EPFL infoscience)
3	Mahmoud, M.S. thesis, Texas A&M; (2024)	Aspen Plus model of a conventional SMR ammonia plant with full stream-by-stream mass-balance tables — the closest thing to a complete citable stream table	Thesis repository (oaktrust.library.tamu.edu)
4	Rice University CENG403 ammonia design projects (1997)	Three coordinated 1000–1016 MTPD designs with ASPEN numbers: reformer (800 °C, 35.3 bar, S/C 3), secondary outlet flows, loop (100 bar, 7.25% purge, 19% per pass)	Free HTML (owlnet.rice.edu)
5	Cheema and Krewer, RSC Advances 8 (2018) 34926	Open-access rigorous Dyson–Simon converter model with all equations and constants; 200 bar / 673–773 K bed design; the kinetics upgrade reference	Open access (rsc.org / pure.mpg.de)
6	Zečević and Bolf, Processes 8(4):408 (2020)	Reformer equilibrium constants (K_{SR1} , K_{WGS}) and approach-to-equilibrium benchmarks, validated on industrial data	Free PDF (mdpi.com)
7	Gillespie and Beattie, Phys. Rev. 36 (1930) 743; tables via El-Gharbawy 2021	The ammonia equilibrium correlation and its pressure-dependent coefficients (Table 4.3), verified in this research	Journal; free via Egypt. J. Chem.
8	Supporting set: Shamiri et al. (2021) plant-validated loop; IFA/Frisse (2004) Uhde plant data; IEA Ammonia Technology Roadmap (2021) energy statistics; Herrmann et al., Fluid Phase Equilibria (2020) loop-binary reference EOS	Cross-checks, energy anchors and the thermodynamic gold standard for the condenser	Mixed free/paywalled

Table 9.1 — Validation library. Sources 1–4 close the mass balance; 5–7 close the converter; 8 closes the condenser and the energy story.

9.2 How to use the library

Sequence matters. Use source 1 first to bound every section during S0–S1 (it exists precisely to describe BAT plants). Use source 4 (free, plain HTML, coarse) as the first whole-train check in S1, then source 2 as the deep loop reference in S2 — its tables cover exactly the variables the tear solver manipulates. Transcribe source 3's stream tables into the golden file when access is secured (it sits behind a CDN block; a library request resolves it). Source 5 is dormant until the kinetics upgrade, source 6 guards the reformer, source 7 ships inside the engine, and source 8's reference EOS defines what "accurate" means for the condenser flash if PR-EOS results are ever disputed. This ordering turns a reading list into a test suite.

9.3 Key web sources for the stack and landscape chapters

For provenance, the load-bearing web sources behind Chapters 3, 5 and 6: DWSIM (dwsim.org) for engine capabilities, licensing and the DWSIM 10 AI assistant; MaximaLabs (maximalabs.io) for browser-simulator pricing tiers; AspenTech's academic program and student-promo pages for the cost wall; LearnChemE (learncheme.com) and Annenberg's Haber interactive for the educational layer; GSE Solutions (gses.com) for OTS practice, installed base and the Nuclearn AI collaboration; OpenAI's realtime, structured-outputs and TTS pricing pages; ElevenLabs, Azure Speech and Deepgram pricing pages for the voice stack; the React Flow documentation (reactflow.dev) for PFD performance practice; and the React Three Fiber guides for the 3D scaling rules. These were all verified live in August 2026; the fast-moving prices should be re-checked before any procurement decision.